



Research Paper

Pear genomes display significant genetic diversity and provide novel insights into the fruit quality traits differentiation

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ABSTRACT

The pear (*Pyrus* spp.) is well known for diverse flavors, textures, and global horticultural importance. However, the genetic diversity responsible for its extensive phenotypic variations remains largely unexplored. Here, we *de novo* assembled and annotated the genomes of the maternal (PsbM) and paternal (PsbF) lines of the hybrid 'Yuluxiang' pear and constructed the pear pangenome of 1.15 Gb by combining these two genomes with five previously published pear genomes representing cultivated and wild germplasm. Using the constructed pangenome, we

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identified 21 224 gene PAVs (Presence-absence variation) and 1 158 812 SNPs (Single Nucleotide Polymorphism) in the non-reference genome that were absent in the PsbM reference genome. Compared with SNP markers, PAV-based analysis provides additional insights into the pear population structure. In addition, some genes associated with pear fruit quality traits have differential occurrence frequencies and differential gene expression between Asian and European populations. Moreover, our analysis of the pear pangenome revealed a mutated SNP and an insertion in the promoter region of the gene *PsbMGH3.1* potentially enhance sepal shedding in ‘Xuehuali’ which is vital for pear quality. *PsbMGH3.1* may play a role in the IAA pathway, contributing to a distinct low-auxin phenotype observed in plants by heterologously over-expressing this gene. This research helps capture the genetic diversity of pear populations and provides genomic resources for accelerating breeding.

Keywords: Pear; Phased diploid genome; Pangenome; PAV; Fruit quality

1. Introduction

The pear (*Pyrus* spp.), an member of the Rosaceae family, holds global renown for its palatable taste, nutritional offerings, and significant economic implications (Su et al., 2022; Xiang et al., 2022; Wang et al., 2023a). Pear is originally from China and has diverse germplasm resources and a long cultivation history. Currently, at least 22 pear species are recognized, with over 5 000 accessions preserved (Wu et al., 2018). This collection spans a myriad of morphologies, a spectrum of physiological differences, and broad adaptability to diverse ecological niches. Within this array of varieties, the ‘Yuluxiang’ pear emerges as a breeding marvel, representing three dedicated breeder generations from the Fruit Research Institute at Shanxi Agricultural University (Wu et al., 2019; Ding et al., 2021; 2024). This pear ascends in popularity, characterized as a mid-maturing, shelf-stable red pear. Its distinguishing traits include a robust size, inviting fruity aroma, vibrant red hue, minimal core, rounded contour, and a succulent, crisp bite. Given these attributes, the ‘Yuluxiang’ pear merits its status as a pivotal genetic material, elucidating its morphology, biology, and genetic nuances.

As bioinformatics assembly algorithms advanced and genome sequencing costs decreased, *Pyrus* genomics studies have proliferated, yielding five distinct genome assemblies from varied pear types. This includes the European cultivar pear (*P. communis*) (Chagne et al., 2014), the Chinese white pear (*P. bretschneideri*) (Wu et al., 2013), a Chinese wild pear (*P. betuleafolia*) (Dong et al., 2020), a Japanese pear (*P. pyrifolia*) (Shirasawa et al., 2021), and a dwarfing pear [(*Pyrus ussuriensis* × *communis*) × spp.] (Ou et al., 2019). Among these, two genomes have undergone assembly and haplotype separation. Yet, the published pear genomes are required to be improved, especially in aspects of genome completeness and contiguity. High-quality pear genomes facilitate studies into the evolutionary history, adaptability, and potential for genetic improvement. Long-read sequencing technology including Oxford Nanopore Technologies (ONT) (Zhang et al., 2023) and PacBio HiFi (Chen et al., 2021; Cheng et al., 2021; Zhou et al., 2024) are required to produce a high-quality genome by enabling the assembly of complex regions, repetitive sequences, and structural variations, leading to an accurate and comprehensive representation of the genome's architecture and functional elements.

Although several genomes from the *Pyrus* genus have been published, the genetic diversity within the genus cannot be sufficiently represented by a single species genome, potentially leaving crucial agronomic traits in non-reference genome regions unexplored. Consequently, constructing a pan-genome is imperative, enabling the encapsulation of the full spectrum of

genetic variations within a species. The dramatic reduction in sequencing costs has facilitated the application of high-throughput sequencing technologies to pan-genomes, as evidenced by the soybean pan-genome (Li et al., 2014). Recently, the pangenomics has expanded from staple crops such as rice (Qin et al., 2021; Wang et al., 2023b) and potato (Tang et al., 2022) to oil crops such as soybean (Liu et al., 2020; Bayer et al., 2021), rapeseed (Song et al., 2020), and sesame (Yu et al., 2019). It also covers horticultural plants such as apple (Sun et al., 2020), tomato (Zhou et al., 2022), and extends to tree species such as *Amborella* (Hu et al., 2022), pecan (Lovell et al., 2021) and yellowhorn (Wang et al., 2023c). The growing emphasis on pan-genomic research is evident and poised to continue (Tahir et al., 2020).

In this context, our study assembled two high-quality pear genomes from the maternal (PsbF) and paternal (PsbM) lines of the hybrid ‘Yuluxiang’ pear, integrating PacBio and Hi-C sequencing datasets. Furthermore, we constructed the pear pangenome by merging the PsbM genome with the PsbF genome and other previously released genomes, offering a comprehensive view of pear genetic resources. Leveraging this newly constructed pangenome, we revisited the short-read sequencing data from 113 varied pear species, and pinpointed a plethora of gene PAVs which might be linked to pear quality traits, revealing differential prevalence between Asian and European pear groups.

2. Materials and methods

2.1. Plant sequencing materials

The prevalent cultivar ‘Yuluxiang’ (*P. sinkiangensis* × *P. bretschneideri*) (Ding et al., 2021) grown in the Fruit Tree Research Institute of Shanxi Agricultural University (37.21° N, 112.30° E) of China was used for doubled haploid *de novo* sequencing. The leaves were used for HiFi sequencing, HiC fixation and library construction, the leaves of ‘Yuluxiang’ parent were used for Illumina sequencing to assist genome assembly, and the mixed collected tissues of leaves, flowers (including pistils and stamens), sepals, stems, and young fruits were used for ONT library construction to assist in predicting the genes of ‘Yuluxiang’ pear. Two weeks after the flowering period, sepal abscission zone from the parents of ‘Yuluxiang’ pear (‘Kuerlexiangli’ and ‘Xuehuali’) were collected for cloning the target fragment of *GH3.1* gene and qRT-PCR.

2.2. DNA and RNA experiment

Total genomic DNA and RNA were extracted from the sepals of the abscission zone of ‘Yuluxiang’ pear and its parent plants, using the CTAB method. Extracted nucleic acids were quantified using a Nanodrop spectrophotometer and assessed for fragment

integrity using an Agilent 2100 Bioanalyzer, to ensure their suitability for subsequent genotype cloning (Table S1).

For ‘Kuerlexiangli’ and ‘Xuehuali’, RNA was isolated from the sepals of the abscission zone at the end of their developmental stage using the CTAB method. This RNA was then reverse transcribed into double-stranded cDNA. The cDNA was used for qRT-PCR according to the previously published research (Ding et al., 2021; Table S2) and gene cloning procedures (Table S3).

2.3. Illumina, ONT, PacBio and Hi-C library construction and sequencing

The qualified 350 bp fragment of DNA was selected to perform pair ended sequencing according to the Illumina HiSeq X ten (150 bp) system standard process (Wright et al., 2017). Qualified RNA from various tissues was extracted and mixed equally for ONT library. Then breaking the RNA into an average of 8 kb fragments was for library construction and ONT sequencing (Leger et al., 2021). The genomic DNA was broken into 15 kb fragments by the Megaruptor® 2 and the SMRT bell library was constructed using SMRT bell Express Template Prep kit 2.0 (Pacific Biosciences), 10 µg of DNA was taken to perform SMRT sequencing on the Sequel II system. After correcting the sequencing error, high fidelity HiFi reads were obtained. More than 2 g of fresh and completely natural condition of growing ‘Yuluxiang’ pear's young branches (cleaned with distilled water in advance) was collected and qualified for Hi-C library. DNA products were obtained through fragmentation, fixation, cell lysis, enzyme digestion, end labeling, flat end connection and purification. Afterwards, 2 µg of DNA samples were taken and interrupted using the Covaris S 220 interrupter. According to the BMK plant Hi-C SOP (BMK190227, Biomarker Technologies, Beijing) reaction system, the DNA ends were repaired, fragments from 300 bp to 700 bp were selected and subjected to quality testing (Qubit3.0 and qPCR were used to detect fragment integrity and concentration). After constructing the library, sequencing was performed using the Illumina platform (PE150).

2.4. De novo genome assembly and Hi-C scaffolding

Based on hifiasm software (Cheng et al., 2021) and incorporating the resequencing data from ‘Yuluxiang’ pear parents, k-mer files for both parent strains were generated. This was achieved using the yak software, with the command `hifiasm -o hifiasm.asm -t 64 -1 R01.yak -2 R02.yak ccs.fastq.gz`. Subsequently, the data were bifurcated into two distinct haplotype genomes: PsbM and PsbF. The subsequent genome scaffolding employed the LACHESIS software (Burton et al., 2013) to facilitate the grouping, ordering and orientation of genomic sequences. Manual mapping and meticulous scrutiny were then performed, culminating in the acquisition of a chromosome-level, doubled genome.

2.5. Genome quality assessment

BUSCO V3 (Simão et al., 2015) with eukaryota_odb9 was used to assess the completeness of the pear assemblies. Centromeric repeats were identified using BLAST+ (version 2.2.3, E-value < 1 e⁻⁵) (Camacho et al., 2009) to search centromere-specific [AGG-GAG]⁵ and telomeric repeats (TTAGGG/CCCTAA) repeats against the genome sequence.

2.6. Repetitive sequence prediction

The assembled ‘Yuluxiang’ pear dual haplotype genome sequences were initially processed using LTR_Harvest (Flynn et al., 2020). Subsequently, the LTR_Finder program (Ellinghaus et al., 2008) was employed for the prediction of long terminal repeat (LTR) sequences. The outputs from the aforementioned software were consolidated using LTR_Retriever (Ou and Jiang, 2018), employing a neutral mutation rate of $\mu = 1.3 \times 10^{-8}$ per base pair per year. Following this, both haplotype genome sequences were subjected to Misa.pl (Ou and Jiang, 2018) and the trf program (Tarailo-Graovac and Chen, 2009). The objective was to forecast microsatellite sequences (SSRs) and tandem repeat sequences (Benson, 1999). This process culminated in the generation of an integrated repetitive sequence library file for the dual haplotype genomes. Finally, the RepeatMasker software (Flynn et al., 2020) was used to process the library file and invoke the `-lib` parameter. This produced the ultimate repetitive sequence, wherein repetitive sequence regions were substituted with ‘N’. Post this, repetitive sequences were excised, streamlining the prediction of the doubled haplotype genes.

2.7. Gene annotation

Extracted high-quality full-length transcripts of ‘Yuluxiang’ pear were used for gene prediction model training based on Nanopore full-length transcriptome data of mixed tissues (leaves, bud, flowers, sepals, stems, and fruits.). The raw sequencing data of Nanopore were filtered and obtained by Pychopper (<https://github.com/friend0/pyChopper>), then the filtered data were treated by the Pinfish package (<https://github.com/nanoporetech/pipeline-pinfish-analysis>), Transdecoder (Perina et al., 2017), maker2 (Holt and Yandell, 2011), AUGUSTUS (Stanke et al., 2006) to train the gene prediction model of ‘Yuluxiang’ pear, and produced a standard .gff format file by EVM (Haas et al., 2008). Finally, the optimal alignment of gene function was inferred using diamond (Buchfink et al., 2015) against KEGG database (Kanehisa et al., 2012), NCBI non redundant (NR) database (Pruitt et al., 2007), EggNOG (Huerta-Cepas et al., 2019), TrEMBL (O'Donovan et al., 2002), InterPro (Mitchell et al., 2015), and SwissProt (Boeckmann et al., 2003) protein databases.

2.8. Non coding RNA prediction

tRNA and rRNA sequences were annotated utilizing tRNAscan-SE (Lowe and Eddy, 1997) and barnap algorithm (v0.9), and both were executed with default parameters. For the identification of miRNAs and snRNAs, the PRfam database (v14.5) (Finn et al., 2014) and the Infernal software (v1.1.1) (Nawrocki and Eddy, 2013) was employed under default settings.

2.9. Structural variations detection

We used the PsbM genome as the reference for SV detection. A European wild pear, a Chinese wild pear, a Xinjiang Pear (PbsF), a Japanese pear, a Chinese white pear and a dwarfing pear were aligned to the PsbM genome, respectively, using Mummer v4.0 (Marcais et al., 2018) with the parameters: `-l 50 -c 100 -maxmatch`. The initial alignment results were filtered using delta-filter with

parameters: -m -i 90 -l 100. The resulting filtered delta files were used as the input for the SyRI pipeline with default parameters (Goel et al., 2019). The detected variations from SyRI comprise genome rearrangements and sequential variations, which can occur in both rearrangement and syntenic regions. Therefore, the sequential variants embedded in the rearrangements were not included for further analysis. According to the definitions of sequence variation in SyRI outputs, we converted these variations into four types of SVs: Insertion (INS and CPG variants), Deletion (DEL and CPL variants), Inversion (INV and INVD variants), and Translocation (TRANS variants). SyRI results were visualized using the plotsr package. SURVIVOR (Jeffares et al., 2017) was used to merge (with parameters: 1000 2 1 1 0 50) the individual genotyped VCF files.

2.10. Pangenome construction

The pear pangenome was constructed using a similar method described in Song et al. (2020) and Tahir et al. (2019). In detail, the PAV sequences in the other six pear genomes relative to the PsbM reference genome were identified using the Mummer 'show-diff' function (Marcais et al., 2018). Initially, sequences intersecting with gap regions within the respective genome were excluded from consideration. Additionally, sequences aligning with gap-start or gap-end boundaries were filtered out. To pinpoint unique sequences within each genome, candidate PAV sequences were aligned to the PsbM genome using minimap2 (Li, 2018) with the parameter setting '-x asm10'. Redundant sequences (PAV sequences showing larger than 80% similarity and coverage with the PsbM genome) were filtered out. The rest of the PAV sequences were retained and linked with 100 Ns bases as the final PAV regions. Genes with at least 80% coding sequence region overlapping with PAV regions were designated as novel genes.

2.11. SNPs calling

Clean reads of 113 pear accessions (Wu et al., 2018) were mapped to the constructed pear pangenome using BWA MEM v0.7.17 (Li, 2013). Default settings were used and duplicates were removed by Picard tools (<http://broadinstitute.github.io/picard/>). Reads were realigned by GATK v3.8-1-0 RealignerTargetCreator and IndelRealigner (McKenna et al., 2010), followed by variant calling using GATK HaplotypeCaller. The resulting SNPs were filtered using criteria including quality by depth (QD) less than 2.0, mapping quality (MQ) less than 40.0, fisher strand (FS) greater than 60.0, QUAL score less than 60.0, MQrankSum less than -12.5, and Read-PosRankSum less than -8.0 to remove low-quality SNPs. High-confidence SNPs were obtained by further filtering out the SNPs with minor allele frequency <0.05 and missing genotype rate >10% using VCFtools (Danecek et al., 2011).

2.12. Gene PAV analysis

Short-read sequencing data of 113 pear accessions were aligned to the constructed pear pangenome using Bowtie2 v2.3.3.1 (-end-to-end - sensitive -I 0 -X 1000) (Langmead and Salzberg, 2012). We used the gene PAV detection approach described in Hu et al. (2020). A gene was classified as absent when the horizontal coverage across exons of the gene was <20% and the vertical coverage was <2×. A gene PAV matrix was generated

using this threshold, with each gene classified as presence or absence for each accession. Statistical significance of PAV frequency changes between European and Asian pear populations was determined by Fisher's exact test. P-values were adjusted for multiple comparisons using the Bonferroni method. PAVs with an adjusted P-value <0.01 and differential occurrence frequency between groups ≥0.2 were identified as under significant occurrence frequency changes between populations.

2.13. Population analysis

The SNP and PAV genotype matrices were used to perform the SNP and PAV based population analysis. Nucleotide diversity values (π) were calculated using Pixy (Korunes and Samuk, 2021) using a 100-kb window. The phylogenetic trees of 113 pear accessions were constructed using IQ-tree using a maximum likelihood method (with the parameters -alrt 1000 -bb 1000) and visualized using ITOL (Letunic and Bork, 2019). Principal component analysis (PCA) was performed using the GAT v1.94 (Yang et al., 2011). The genetic differentiation (F_{st}) between sub-populations was computed using a 100-kb sliding window and a step size of 10 kb with VCFtools (v0.1.15) (Danecek et al., 2011). The population structure analysis was performed using the ADMIXTURE V1.3.0 (Alexander et al., 2009).

2.14. GO analysis

Gene function annotation was performed using Blast2GO v2.5 (Conesa et al., 2005). Genes in the pangenome were aligned to the proteins in the Viridiplantae database using BLASTP (E-values < 1×10^{-5}) (Camacho et al., 2009). Gene ontology (GO) analysis was conducted using topGO (Raudvere et al., 2019) and Fisher's exact test with 'elim' used to correct for multiple comparisons.

2.15. RNA-seq differential expression analysis

Clean RNA sequencing reads (Zhang et al., 2016) were generated by removing low-quality reads, adapters, and reads containing poly-N using the fastp v0.20 (Chen et al., 2018) with default parameters. Clean reads were further mapped to the constructed pear pangenome using Hisat2 v2.1.0 (Pertea et al., 2016) with default parameters, and the number of mapped reads was counted using HTSeq v0.6.1 (Anders et al., 2015). RNA-Seq reads were normalized to TPM (Transcripts Per Kilobase Million) and low expression reads (TPM < 1) were removed. Differential expression analysis was performed using the DESeq2 R package 1.18.0 (Love et al., 2014) between the European pear population (RC: Red Clapp's Favorite mutant, *P. communis*) and the Asian pear populations (HS: 'Hosui', *P. pyrifolia*; YL: 'Yali', *P. bretschneideri*; KRLXL: 'Kuerlexiangli', *P. sinkiangensis*; NG: 'Nanguo', *P. ussuriensis*).

The raw genome sequenced data pertinent to this research have been deposited in the National Genomics Data Center, China National Center for Bioinformation/Beijing Institute of Genomics, Chinese Academy of Sciences (GSA: PRJCA018905 and CRA012175) and are publicly accessible at NGDC. The reported resequencing of 113 pear accessions from worldwide and RNA sequencing of pear fruit in five cultivars are detailed in Wu et al. (2018) and Zhang et al. (2016) in PRJNA381668 and PRJNA309745, separately.

'Yuluxiang' pear double haploid genomes and pear pan-genomes annotation can be accessed at [10.6084/m9.figshare.23950353](https://doi.org/10.6084/m9.figshare.23950353).

2.16. Functional validation of the heterologous overexpression

Sterilization was carried out on rice (*Oryza sativa* L. Nipponbare) seeds at the grain-filling stage using the following steps. Seeds with full grains were selected and then rinsed thoroughly with clean water. The selected seeds were placed in a beaker and sterilized using 70% alcohol for 2 min. Subsequently, a 30% NaClO solution was added until the beaker was filled, and the seeds were left in the solution for 90 min. After discarding the NaClO solution, the sterilized seeds were rinsed at least four times with deionized water (ddH₂O) to remove any remaining surface residues.

2.17. Gene cloning, quantification, and transgenic materials

The abscission zone tissues of the 'Yuluxiang' pear and its parents of 'Kuerlexiangli' (*P. sinkiangensis*, represented maternal haplotype genome PsbF) and 'Xuehuali' (*P. bretschneideri*, represented paternal haplotype genome PsbM) were collected at the end of sepal development (two weeks after blooming) in the Fruit Tree Research Institute of Shanxi Agricultural University (112°48' E, 37°38' N). Seeds at the grain-filling stage were used in rice (*O. sativa* L. Nipponbare).

2.18. Construction of overexpressed vector of the pEGOEP35S-H-PsbMGH3.1-GFP

According to the cloning primers, after PCR amplification of target fragments, ligation-mediated restriction, recombinant plasmid transformation and colony plasmid extraction, the quality of the extracted plasmids was measured by agarose gel electrophoresis at a concentration of 1%. Finally, Sanger sequencing was performed based on the primer eGFP-cx, and the target genes were aligned with the *de novo* assembled doubled haploid genomes.

2.19. Callus induction and subculture

Using sterilized tweezers, the pre-treated seeds were handled carefully. Young rice embryos were then isolated using sterilized steel hooks and placed on sterile filter paper. These embryos were transferred onto an induction culture medium. Afterward, the Petri dish was sealed with plastic wrap and incubated under dark conditions at 25 °C for five days. After that the embryogenic callus tissues were selected for further subculturing over a three-day period. Subsequently, the constructed *Agrobacterium* vectors were co-cultured with the induced tissue and subjected to antibiotic screening, induced differentiation, and rooting steps to obtain a preliminary positive transgenic seedling (Tie et al., 2012). The method of transgenic tomato was performed as the previous research (Guo et al., 2024).

2.20. Molecular docking

The crystal structure of the VvGH3.1 protein from grape (PDB ID: 4B2G) was obtained from the Protein Data Bank (PDB) database (<https://www.rcsb.org>). Similarly, the chemical structure of IAA (PubChem ID: 802) was downloaded from the PubChem database. These structures were then used in docking studies conducted by

Autodock (Li et al., 2022b). The study involved docking IAA with the protein encoded by the gene PsbMGH3.1 (Peat et al., 2012). Among the various conformations generated during docking, the one demonstrating the highest binding affinity was selected as the final docking conformation.

3. Results

3.1. Chromosome-length assemblies of pear genomes

Using the PacBio CCS (HiFi) reads and Illumina short reads, we successfully *de novo* assembled two parent genomes of the hybrid 'Yuluxiang' pear cultivar from Jinzhong city, Shanxi, including the maternal line PsbF (*P. sinkiangensis*) from Korla city, Xinjiang, and the paternal line PsbM (*P. bretschneideri*) from Shijiazhuang city, Hebei, China (Fig. 1, A). Following correction for misjoins, the contigs were ordered, oriented and anchored to chromosomes using Hi-C sequencing (Fig. 1, B–D). This allowed the anchoring of 96.2% and 98.8% of sequences to chromosomes to PsbF and PsbM, respectively. Both genome assemblies demonstrated near gap-less qualities, with ten of seventeen chromosomes comprising only one contig and the remaining chromosomes containing fewer than three contigs (Table S4). Centromere and telomere-specific repeats were identified across almost all chromosomes in both the PsbF and PsbM genomes (Table S5), indicative of the high-quality assembly of these genomes. Moreover, at least 44.5% of repeat sequences were identified within the assembled PsbF and PsbM genomes, with long terminal repeat (LTR) retrotransposon elements being the most prevalent class of transposable elements (TEs), accounted for at least half of the TEs (Table S6). The assembly lengths of the PsbF and PsbM genomes are around 538 and 510 Mb, respectively. Both genome assemblies have a contig N50 over 26.6 Mb, which is higher than previously published pear genomes (Fig. S1). Benchmarking Universal Single-Copy Orthologs (BUSCO) and Cluster of Essential Genes (CEG) evaluation confirmed the completeness of the PsbF and PsbM genomes to exceed 96% and 98% (Table S7). These results illustrate a high level of contiguity and completeness of our assembled pear genomes.

Through a combination of RNA-seq transcript mapping, *ab initio* prediction, and homologous protein searches, we annotated 46 014 and 45 317 gene models in the PsbF and PsbM assemblies, respectively (Fig. S1), wherein around 90% of genes garnered functional annotation from at least one functional protein database (Table S8). When examined from a genome-wide perspective, gene models exhibited greater density in chromosome arms with fewer repeat elements. Additionally, we annotated 8 514 and 7 402 non-coding RNAs, including rRNAs, tRNAs, miRNAs, and snRNAs, in the PsbF and PsbM genomes, respectively (Table S9).

3.2. Large genomic variations in seven pear genomes

The high-quality pear enables the discovery of structural variations (SVs) that are equal to or larger than 50 bp in size. In the PsbM genome, we identified SVs in six other pear genomes including a European wild pear (*P. communis*, Pco) (Chagne et al., 2014), a Chinese white pear (*P. bretschneideri*) (Wu et al., 2013), a Chinese wild pear (*P. betuleafolia*, Pbe) (Dong et al., 2020), a Japanese pear (*P. pyrifolia*, Ppy) (Shirasawa et al., 2021), and a dwarfing

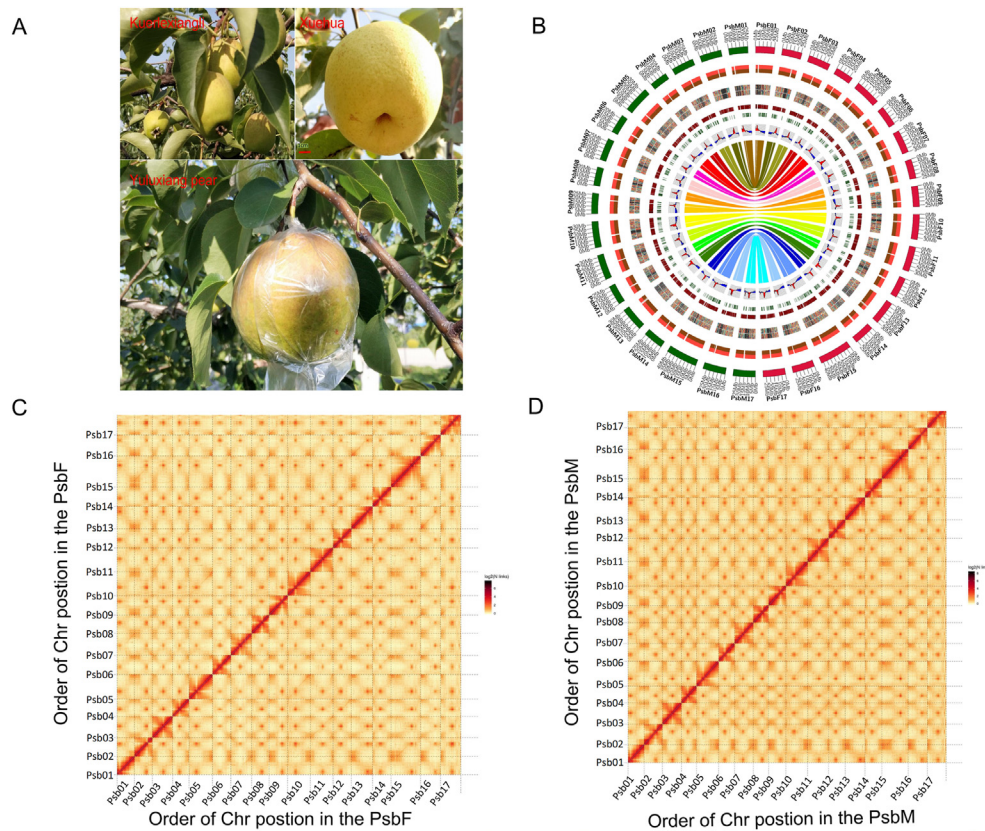


Fig. 1 Genome assembly of PsbM and PsbF pear genomes

(A) ‘Yuluxiang’ pear and its maternal ‘Kuerlexiangli’ (PsbF) and paternal ‘Xuehuali’ (PsbM). Red bar means 1 cm. (B) Genome-wide characteristics of the PsbM and PsbF pear genomes. From outside to inside denotes chromosome, protein coding gene (red represents positive chain, brown represents negative chain), rRNA (5.8S, 5S, 18S, 28S), tRNA, repeat density (500 kb sli.00000000 window, the proportion of repeat sequences per sliding window-the average proportion of all slide window repeat sequences, red for those above the average, blue for those below the average), and allele connection, respectively. (C, D) Chromosome contact maps across the seventeen chromosome-length scaffolds of the PsbF (C) and PsbM (D) pear genomes, respectively. The abscissa and ordinate represent the order of each bin on the corresponding chromosome group.

pear [(*P. ussuriensis* × *communis*) × spp., Pdr] (Ou et al., 2019). In total, 44 681 non-redundant SVs, including 247 inversions, 814 translocations, 431 duplications, 14 837 deletions, and 27 979 insertions were identified, with translocations and inversions having longer sequence lengths than deletions and insertions (Fig. 2, A). Interestingly, compared with other pear genomes, Pdr and Ppy contained the largest number of SVs (Table S10). Furthermore, we detected 6 341 high-confident SVs across at least two genomes, encompassing 7 inversions, 15 translocations, 3 989 deletions, and 2 330 insertions (Table S11). We also found 319 high-confident SVs located in the gene exon regions that will potentially be affecting the gene functions (Table S12).

Previous research has highlighted the link between inversions and the suppression of local recombination (Crow et al., 2020; Hamala et al., 2021; Hu et al., 2024a). An example of this is an inversion covering 646 kb on chromosome 13 (PsbM13: 19 118 031–19 764 319), identified exclusively in the genomes of Ppy and Pdr, but not in those of PsbM, PsbF, Pco, Pbr, and Pbe (Fig. 2, B). Within this inversion in the PsbM genome, there are 27 gene models, notably including two genes (PsbM013G02206 and

PsbM013G02207) associated with chromosome partitioning. These genes are located near the inversion's distal breakpoint, approximately 119 Kb away, suggesting their potential role in influencing recombination suppression in this region (Table S12). Moreover, deletions and insertions were reported to be related to adaptation to varying pathogen pressures within different growing environments (Zmienko et al., 2014; Dolatabadian et al., 2022; Hu et al., 2022, 2024b). In this study, we identified that 2 insertions and 9 deletions were overlapped with the exon regions of genes which were functionally associated with disease resistance (Table S12). For example, a 663 bp deletion leading to three exons absent was identified in the gene PsbM006G00203 (homology of *Arabidopsis* disease resistance protein RPM1) and a 9 393 bp deletion resulting in complete loss of two *Arabidopsis* disease resistance protein homologous genes (PsbM015G01503 and PsbM015G01504) among the Pco, Pdr, and PsbF genomes. These findings underscore the significant impact of structural variations on the genomic functionality of disease resistance mechanisms in these species, suggesting potential vulnerabilities or adaptations in their response to pathogens.

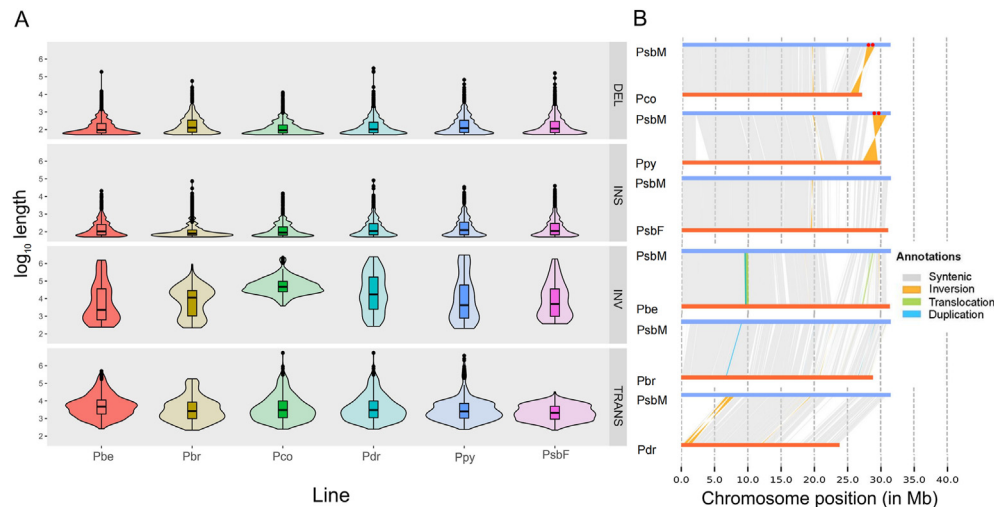


Fig. 2 Structural variations across seven pear genome assemblies

(A) The size of SVs (insertions, deletions, translocations and inversions) in six genomes compared to the PsbM genome. (B) Synteny map showing the structure of Chromosome 13. Red dots between 25 Mb and 30 Mb show the position of two proline-serine-threonine phosphatase-interacting protein genes (*PsbM013G02206* and *PsbM013G02207*) functionally associated with cell division.

3.3. Construction of the pear pangenome

Using the pangenome construction method similar to the pangenomics studies of *B. napus* (Song et al., 2020) and *B. oleracea* (Golicz et al., 2016), we constructed the linear pear pangenome by adding the PAV sequences from other six pear genomes to the PsbM reference genome. The constructed pear pangenome is 1.15 Gb in size, containing an additional 635 Mb non-reference sequences and hosting 21 560 newly annotated high confidence genes which were missing in the reference genome. A higher percentage (76.0%) of repetitive elements accounted for the non-reference genome than the reference genome (54.3%) (Table S13), indicating that transposable elements might be the important driver of presence/absence variations. By analyzing the read mapping of short-read sequencing data, we found that the read mapping rate aligned to the constructed pear pangenome was higher than that aligned to the PsbM reference genome (Fig. 3, A). The pear pangenome shows a further improvement in BUSCOs evaluation with a higher complete BUSCOs score (93.5%) and a lower missing (3.2%) and fragmented (3.3%) BUSCOs than the PsbM genome (complete BUSCOs: 92.9%; missing BUSCOs: 3.4% and fragmented BUSCOs: 3.4%) (Table S14). These findings demonstrate that compared with a single reference genome, the pear pangenome can reduce the reference bias and capture the missing genetic diversity. Moreover, pangenome modeling revealed a closed pan-genome, suggesting our constructed pear pangenome can capture nearly all the pear gene contents (Fig. 3, B).

To detect the population-wide SNPs and PAVs in the pear pangenome, we further aligned 113 pear resequencing data (Wu et al., 2018) to the pear pangenome in which these pear accessions were from worldwide collections representing the genetic diversity of cultivated and wild pear species (Table S15). We identified 18 667 069 SNPs with 1 158 812 (5.4%) SNPs located in the non-reference sequences and further performed the SNP variant annotation to assess their functional effect. Subsequently, SNPs located in non-reference sequences were more likely to have significant impacts on gene function, with 59.3% and 2.7% of the SNPs

located in non-reference sequences and a lower percentage of SNPs (51.2% and 0.9%) located in reference sequences resulting in missense and nonsense mutations respectively. In addition, 326 461 PAVs ranging from 100 bp to 832.3 kb were identified across 113 pear accessions using the pear pangenome. Based on the PAVs detection result, 32 174 (48.4%) were core genes and present in all 113 pear accessions, while 51.6% (34 367) of the genes were classified as dispensable, comprising 9 624 softcore, 22 885 shell, and 1 858 cloud genes, which were defined with a present frequency of more than 99%, 1%–99%, and less than 1% respectively (Fig. 3, C). The proportion of dispensable genes in this study is relatively higher compared with other fruits such as apple (18.7%) (Sun et al., 2020), tomato (26%) (Gao et al., 2019), and watermelon (30.5%) (Wu et al., 2023). However, the proportion is similar to cucumber (43.8%) (Li et al., 2022a) and smaller than citrus (64.6%) (Gao et al., 2023). Gene ontology (GO) enrichment analysis shows that genes with essential biological functions, including DNA-templated transcription, transmembrane transport and response to potassium ion were enriched in core genes (Table S16). By contrast, genes with functions in pollen recognition, regulation of photorespiration, biotic stimulus response, and disease resistance are enriched in dispensable genes (Fig. S2; Table S17). Genes in dispensable orthologous clusters have shorter gene lengths, lower gene expression levels and higher non-synonymous/synonymous substitution ratio (*Ka/Ks*) (Fig. 3, D–F). In line with the cucumber pangenome study (Li et al., 2022a), our results suggest dispensable genes are under stronger diversifying selection and have a faster evolutionary rate than core genes, and may be associated with phenological changes and disease resistance variations during selection for adaption to different environments.

3.4. Characterization of the population using the pear pangenome

We further used our pangenome to characterize the pear population. Calculation based on SNPs makers, the nucleotide diversity (π) of pear at the pangenome level across 113 pear

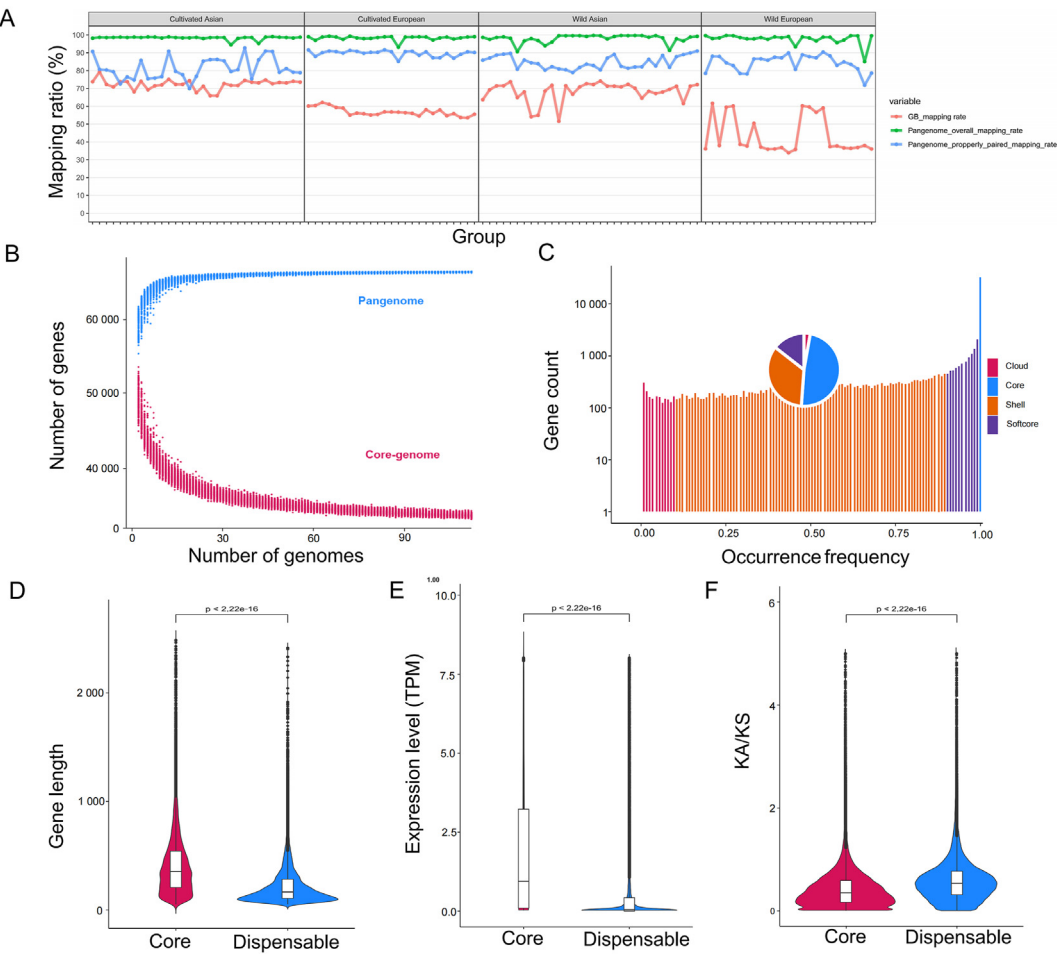


Fig. 3 Pangenome of pear

(A). Mapping ratio of 113 pear resources based on pangenome and single genome. Mapping rate denoted (pink line) mapping rate based on Wu et al. (2018). (B) Pan-genome modeling. The pan-genome modeling shows no more dramatic increases when the number of accession genomes is over 30, indicating that the pear pangenome is sufficient to capture the majority of PAVs within pears. The upper and lower lines represent the pan-genome number and core-genome number, respectively. (C) Pangenome gene classification. (D) Boxplots show the median and upper and lower quartiles of gene length. (E) The RNA-seq expression level. (F) Non-synonymous/synonymous substitution ratio (Ka/Ks) of core and dispensable orthologous gene clusters.

accessions (Wu et al., 2018) was 4.84×10^{-3} . Similar to Wu et al. (2018), the cultivated Asian and European pears have similar levels of nucleotide diversity (3.56×10^{-3}) but lower than the wild Asian (4.23×10^{-3}) and European pears (3.92×10^{-3}) (Fig. 4, A). Using PAVs as markers, the PAV diversity of cultivated (1.97×10^{-3}) and wild (2.42×10^{-3}) European pears was lower than cultivated (2.77×10^{-3}) and wild (2.88×10^{-3}) Asian pears (Fig. 4, B), suggesting that European pears contained fewer PAVs than the Asian pears at the pangenome level. Moreover, levels of population fixation index (FST) were estimated using the pangenome. Both SNPs and PAVs based analysis show a similar population differentiation with a similar FST between different pear populations (Fig. S3, A, B). In particular, large population differentiation was observed between the Asian and European pears, with an average FST over 0.28 between cultivated and wild pear species. Using the pear pangenome, we further studied the population structure of 113 diverse pears. The cultivated and wild Asian and European pears can be separated into four clusters at both the SNPs based and

PAVs based PCA (Fig. 4, C and D) and phylogeny results (Fig. 4, E and F). PAVs based PCA can better distinguish the cultivated and wild European pears (Fig. 4, D). In the PAV-based phylogeny structure, wild Asian pears *P. calleryana*, *P. dimorphophylla*, and *P. phaeocarpa* show a closer phylogenetic relationship with cultivated Asian pear *P. pyrifolia* (Fig. 4, F), which is not reflected in the SNP-based phylogeny structure (Fig. 4, D). We also performed the genetic ancestral components analysis to study the population structure of pears based on the SNPs (Fig. S4) and PAVs markers (Fig. S5). Both PAVs and SNPs based analysis showed that when $K = 4$, the pear populations can be separated into four groups, including cultivated and wild Asian and European pears. SNPs based population structure analysis has the lowest the coefficient of variation (CV) when $K = 5$, which further divided the Asian wild pears into two groups. However, PAVs based population structure analysis has the lowest CV when $K = 3$. In this case, cultivated and wild European pears were clustered into one group. PAVs-based analysis at the pangenome level (Amas et al., 2023)

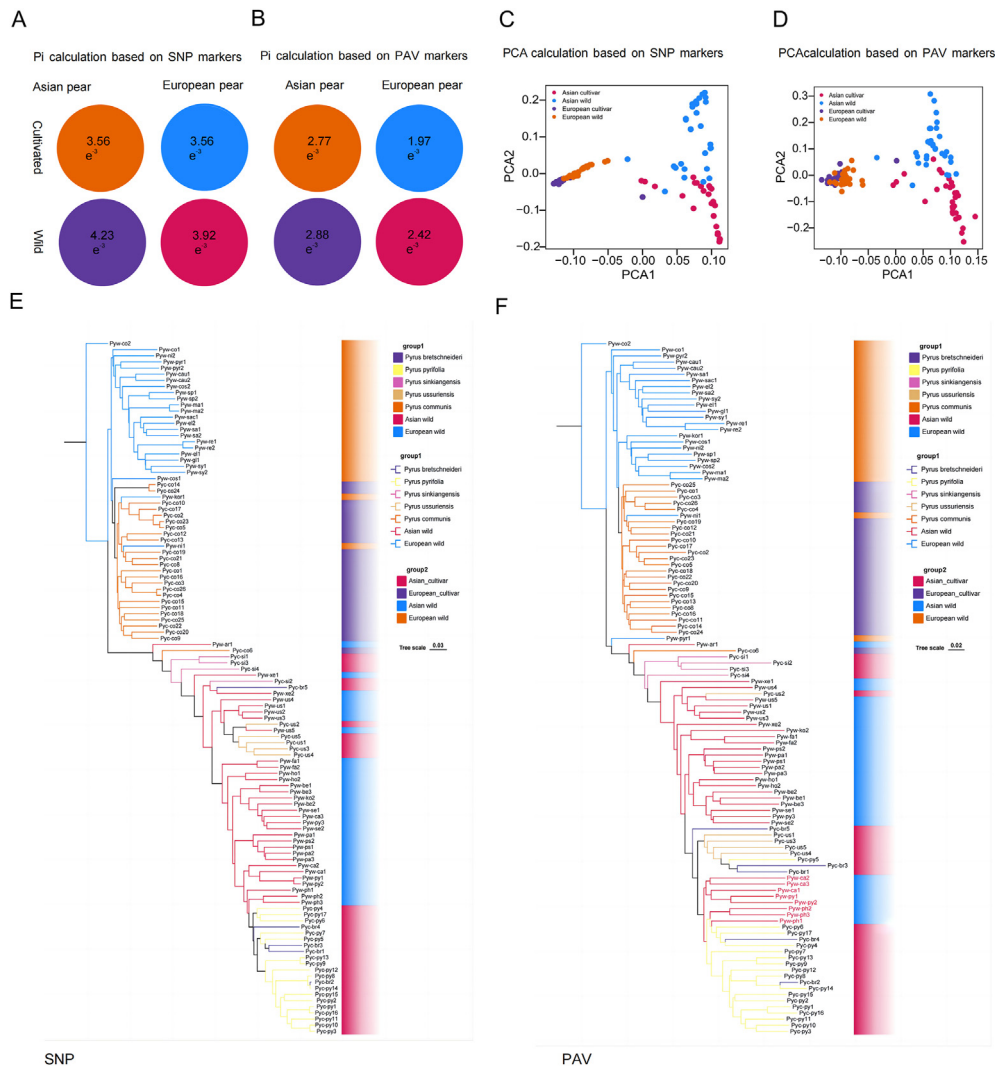


Fig. 4 Characterization of the pear population basing on the pear pangenome

(A) and (B) represented the calculation of nucleotide diversity based on SNPs and PAVs. Nucleotide diversity (π) across 113 pear accessions at the pangenome level was 4.84×10^{-3} , with cultivated Asian and European pears showing similar levels (3.56×10^{-3}), which are lower than their wild species, while PAV diversity indicated European pears (both cultivated and wild) have lower diversity than Asian pears. (C) and (D) represented the PCA analysis based on SNPs and PAVs reveals different patterns in the pear population structure. (E) and (F) denoted the phylogenetic trees constructed based on SNPs and PAVs reveal different patterns in the pear population structure.

provides additional insight into the characterization of population structure in pears.

3.5. Gene PAVs provide new insight into the selection footprints of European and Asian pears

Analyzing gene content across diverse pear accessions demonstrated a significant difference in average gene number per individual between European pear and Asian pear. European pear contains a higher average number of genes (56 842), with a reduction in Asian pear (56 118) (Fig. 5, A). Different from previous findings that gene number reduction was observed during crop domestication and breeding improvement (Gao et al., 2019), cultivated European pear and Asian pear do not have significantly different gene numbers between wild European pear and wild

Asian pear respectively (Fig. S6). The reduction in average gene numbers hides a more complex pattern of increases and decreases in the frequency of specific genes across the population. To identify gene PAV changes across diverse pear accessions, we compared gene occurrence frequencies between European pear and Asian pear, and between their wild and cultivated individuals (Fig. 5, B). A total of 8 361 genes showed a higher occurrence frequency in European pears (Table S18), while 8 093 genes showed an increased occurrence frequency in Asian pears (Table S19). GO enrichment analysis revealed that genes showing a higher occurrence frequency in Asian pears are associated with the function of defense response, response to biotic stimulus, vessel member cell differentiation, and sterol biosynthesis (Table S20).

To further investigate how PAVs affect fruit quality in pear, we identified genes showing significant differences in PAV

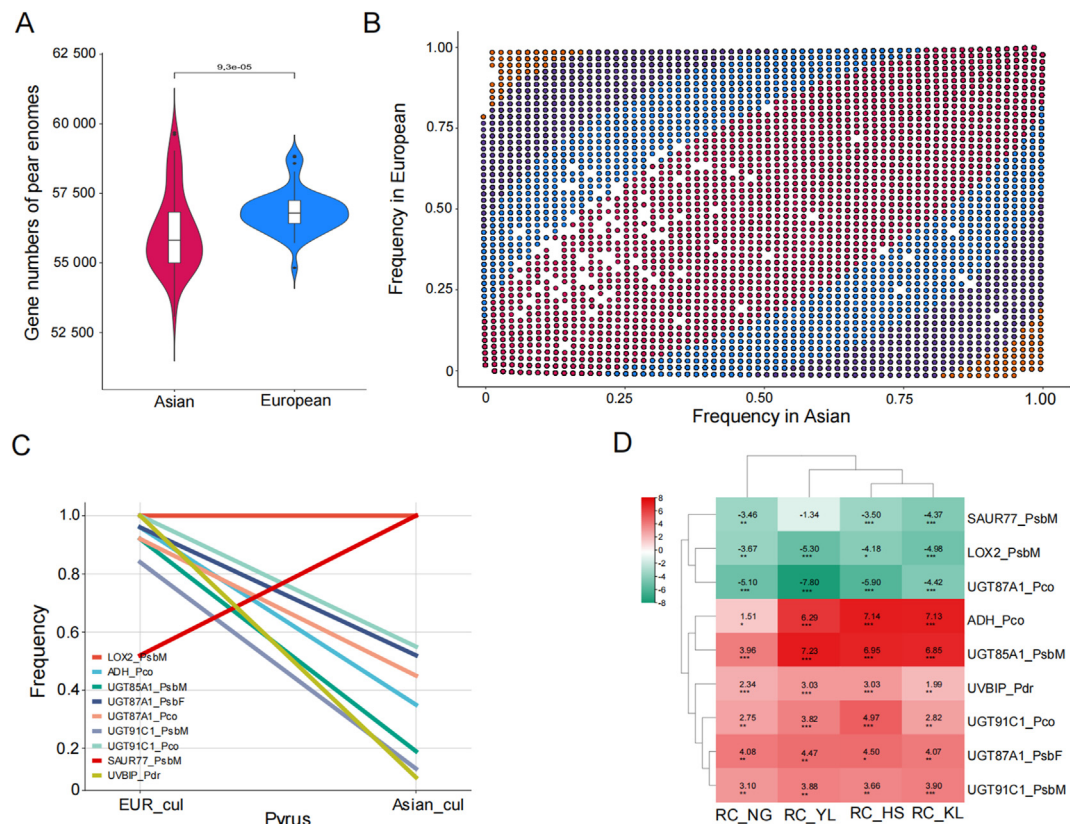


Fig. 5 Gene PAVs between European and Asian pears

(A) The gene number between Asian and European pear genomes. (B) Change of gene PAV occurrence frequency between Asian and European pear genomes. Colors indicate difference P-value of the gene occurrence frequency between European and Asian pears, with orange ($P < 1e^{-20}$), purple ($1e^{-20} < P < 1e^{-10}$), blue ($1e^{-10} \leq P < 0.01$), and red ($P \geq 0.01$). (C) The change of PAV occurrence frequency of nine genes between Asian and European pear genomes. These nine genes may play key roles in fruit quality trait differentiation between Asian and European pear genomes. (D) The differentiation of gene expression among nine key genes between Asian and European pear genomes.

occurrence frequency between Asian and European pear populations that are functionally associated with fruit shape, aroma, color, stone cells, and fruit hardness. In total, we identified 192 fruit quality traits associated genes (Table S21), with homologous six genes exhibiting significant correlations with fruit shape development, including transcription repressor gene OFP16 (*PsbM011G02493* and *PsbM005G03117*) (Snouffer et al., 2020), homeobox protein knotted-1-like gene KNOX1 (*Pco_Presence_Seq_006877*) (Wang et al., 2022; Sun et al., 2023), SUN domain-containing protein 5 gene (*Pdr_Presence_Seq_013651*) (Ma et al., 2022), and TRM (*PsbM003G00721* and *Pco_Presence_Seq_003059*) (Zhang et al., 2023). For example, the gene *PsbM005G03117* is present in all Asian pear accessions but less than half of European pear accessions (Occurrence frequency: 0.55) (Differential occurrence frequency: 0.45, P-value: 1.54×10^{-8}), while *PsbM011G02493* showing a different pattern between present in all European pear accessions and near three-quarters of Asian pear accessions. The largest differential occurrence frequency (0.67) between Asian and European pears is observed in the gene *SUN5* (*Pdr_Presence_Seq_013651*) (Occurrence frequency in Asian and European pear: 0.33 and 1, P-value: 1.56×10^{-14}). Besides, we identified 28 genes associated with fruit color

development, including 12 LOB domain-containing protein (LBD) genes, 10 B-box zinc finger protein (BBX) genes (Bu et al., 2022) and 4 anthocyanin regulatory (Myb_DNA-binding) protein genes (Jin et al., 2016). Among these genes, a LOB domain-containing protein CRL gene (*PsbM001G01311*) shows a significantly lower occurrence frequency in European pear, especially in cultivated European pear compared with Asian pear (Differential occurrence frequency between cultivated Asian and European pears: 0.88, P-value: 5.00×10^{-11}), while a B-box zinc finger protein 22 gene (*Pdr_Presence_Seq_002427*) is present in all cultivated European pear but nearly absent in cultivated Asian pear (Differential occurrence frequency between cultivated Asian and European pears: 0.96, P-value: 8.40×10^{-13}). Moreover, 57, 3, and 102 genes playing important roles in the development of fruit stone cells, fruit aroma, and firmness show differences in PAV occurrence frequency. Additionally, the UV-B-induced protein gene *UVBIP* (*Pdr_Presence_Seq_004483*) (Qian et al., 2014) regulating photomorphogenic responses to UV-B and blue light shows higher occurrence frequency in European pear than in Asian pears (Differential occurrence frequency between cultivated Asian and European pears: 0.92, P-value: 2.54×10^{-24}) (Fig. 5, C).

We further performed gene expression analysis merging RNA-seq data from five pear species and pear fruit tissues of seven stages (Zhang et al., 2016). Subsequently, we identified six fruit firmness and two aroma-related genes showing significant PAV occurrence frequency as well as significant differential gene expression between Asian and European pear populations.

Compared with the Asian pear, the fruit hardness development genes including the UDP-glycosyltransferase gene UGT87A1 (Wu et al., 2017; Cao et al., 2023) (Pco_Presence_Seq_002869, Differential occurrence frequency between cultivated Asian and European pears: 0.43, P-value: 1.75×10^{-5}) and the auxin-responsive protein gene SAUR (Zhang et al., 2022) (PsbM005G02020,

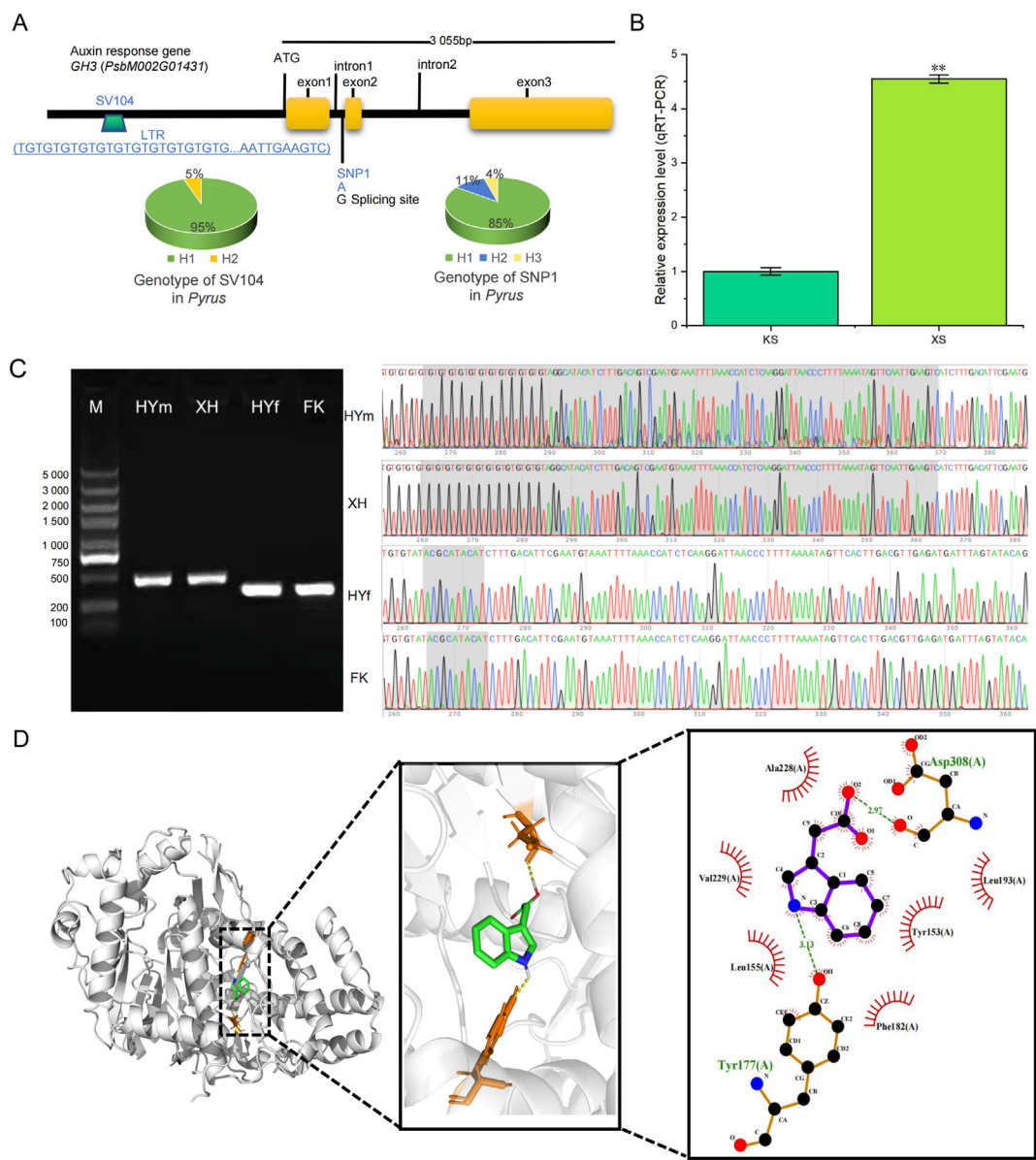


Fig. 6 The characterization of the pear sepal abscission zone gene GH3.1 from structural variation, expression to functional validation

(A) The variation of gene GH3.1 in *Pyrus*. (B) Expression of gene PsbMGH3.1 in exfoliated XS and persistent KS sepals zone of parents of 'Yuluxiang' pear. XS and KS represent the late stage of sepal development of 'Xuehuali' and 'Kuerlexiangli' respectively. $n = 3$, T test, ** denotes $P < 0.01$. (C) Cloning and genotyping of gene GH3.1 at insertion sites of 104 bp between 'Yuluxiang' pear and its parents by electropherogram. The left part (Electropherogram): The inserted mutation sites of 'Yuluxiang' pear and its parents. M: Marker. HYm and XH, and HYf and FK, represent the cloning fragments of the allele PsbMGH3.1 from upstream to downstream of the mutation sites of 'Yuluxiang' pear and 'Xuehuali', respectively, and the cloning fragments of the allele PsbFGH3.1 from upstream to downstream of 'Yuluxiang' pear and 'Kuerlexiangli'; The right part (Sequencing map): Detection of genotyping of mutation sites. (D) Protein PsbMGH3.1 binds directly to IAA through hydrogen bond. The left part: Simulation of molecular docking between PsbMGH3.1 and IAA, and local amplification of active pocket; The middle part: IAA chemical structure formula; The right part: Hydrogen bond force between PsbMGH3.1 and IAA.

Differential occurrence frequency between cultivated Asian and European pears: -0.34 , P -value: 5.76×10^{-7}) show a significant downregulation in expression in European pear fruit ($\log_2$ FoldChange > 3.46). However, the other four UDP-glycosyltransferase genes (*Pco_Presence_Seq_001484*, *PsbF_Presence_Seq_007967*, *PsbM015G01882* and *PsbM017G01217*) having a higher occurrence frequency in European pear (Differential occurrence frequency between cultivated Asian and European pears > 0.22 , P -value $< 5.26 \times 10^{-4}$), showed an upregulated expression in European pear ($\log_2$ FoldChange > 2). Among these genes, *PsbF_Presence_Seq_007967*, a homolog of *Pbr011574*, was reported to be located in the fruit firmness QTL according to a previous study (Wu et al., 2018). In addition, two aroma-related genes, alcohol dehydrogenase (ADH) (Zhang et al., 2010; Shi et al., 2019) and Lipoxygenase 2 gene (LOX) (Luo et al., 2021; Leng et al., 2023), also show significant differential gene expression between Asian and European pear populations. For example, the ADH gene (*Pco_Presence_Seq_011843*) displays a higher occurrence frequency (Differential occurrence frequency between cultivated Asian and European pears: 0.45 , P -value: 3.17×10^{-6}) as well as an upregulated expression in European pear ($\log_2$ FoldChange > 1.51). Furthermore, the UVBIP gene (*Pdr_Presence_Seq_004483*) with a higher occurrence frequency in European pear shows a significantly elevated expression in European pear compared to Asian pear (Fig. 5, D; Table S22). The results suggest that gene PAVs might play key roles in differentiating pear quality traits between Asian and European pears by regulating the expression of related genes.

3.6. Exploring structural variation, expression, and functional validation of the sepal abscission zone pear gene GH3.1

Based on previous studies (Qi et al., 2013; Guo et al., 2024; Huang et al., 2023), GH3.1 (*PsbM002G01431*), encodes an indole-3-acetate synthetase which is a pear sepal development candidate gene and plays an important role in early auxin response. Analysis of allele-specific structural variations within the gene GH3.1 promoter unveiled a 104 bp insertion (SV104) in the genome of *PsbM* compared to the genome of *PsbF*, spanning from position 11 287 633 bp to 11 407 583 bp on the upstream promoter of GH3.1 (Fig. 6, A). Based on the genomes of seven published species, wild pear (*P. betuleafolia*, Dong et al., 2020) and *PsbM* exhibit natural sepal shedding, while the other 5 varieties [*P. communis*, Chagne et al., 2014; *P. bretschneideri*, Wu et al., 2013; *P. pyrifolia*, Shirasawa et al., 2021; (*P. ussuriensis* $\times$ *communis*) $\times$ spp., Ou et al., 2019; *PsbF* assembled in this study]; exhibit sepal persistence in fruit sepals. In addition, an SNP1 (G $\rightarrow$ A) in the first intron of the gene GH3.1 potentially affecting gene alternative splicing was identified in *PsbM* (Fig. 6, A) and *Pbe* (mature fruit with completely shedding of sepals) (Fig. S7). By analyzing the genotype of GH3.1 gene mutation sites in 7 pear genomes, we speculate that the variable splicing sites of GH3.1 genes, especially those located in introns and exons, may be related to sepal shedding. We conducted a further examination of the distribution of both the insertion and SNP in 113 pear accessions. We identified five accessions (4 from cultivated and wild Asian pears, namely *Pyc_br1*, *Pyc_br5*, *Pyw_us5*, *Pyw_xe1*, 1 from wild European pear, namely *Pyw_cau1*) (Table S23) that harbored the same SNP as *PsbM*, while an additional six accessions (four from Asian

cultivated pears *Pyc_py5*, *Pyc_py7*, *Pyc_si1*, *Pyc_us1*, two from wild pears *Pyw_pa1* and *Pyw_pa3*) (Table S24) exhibited the insertion. Furthermore, the expression of *PsbMGH3.1* in the sepal abscission zone of the paternal haplotype genome *PsbM* ('Xuehuali') was five times higher than that in the maternal haplotype genome *PsbF* ('Kuerlexiangli', mature fruit with incompletely shedding of sepals) (Fig. 6, B and Table S25). Notably, these shared genetic variants aligned precisely with the paternal genotype of the 'Yuluxiang' pear haploid genome, providing insights into the genetic diversity of this distinct lineage.

We further designed primers based on the flanking sequences of the insertion for validation, confirming the precise alignment of cloned variant sites in the 'Yuluxiang' pear genome and its parental genome with those in the assembled doubled haploid genomes (*PsbF* and *PsbM*) (Fig. 6, C and Fig. S8). Utilizing the protein crystal structure of VvGH3.1 (Peat et al., 2012; Wang et al., 2015) from grape (PDB ID: 4B2G) based on homology (Fig. S9) and IAA (PubChem ID: 802), we constructed a molecular docking model, revealing the active pocket and binding site predictions of *PsbMGH3.1* with IAA. Among nine identified binding modes, the final docking conformation with an optimal binding energy of $-7.0 \text{ kcal} \cdot \text{mol}^{-1}$ was selected (Table S26 and Fig. 6, D). The interactions between IAA and *PsbMGH3.1* were primarily characterized by hydrogen bonding with Asp308 and Tyr177, with spatial distances of 2.97 Å and 3.13 Å, respectively. Notably, overexpression of the sepal shedding gene *PsbMGH3.1* in rice led to noticeable alterations in transgenic plant phenotypes, including significantly reduced adventitious root length, the longest leaf length, and plant height compared to wild plants (Table S27; Figs. S10 and S11). The phenotype observed in 'Proto' (*Solanum lycopersicum* L.) and rice expressing transgenic *PbGH3.1*, both displayed a low-auxin phenotype characterized by dwarfism (Fig. S12).

4. Discussion

We *de novo* assembled and annotated the genomes of the maternal (*PsbM*) and paternal (*PsbF*) lines of the hybrid 'Yuluxiang' pear using the PacBio long read and Hi-C sequencing. Compared with other published pear genomes, our genome assemblies show significant completeness and accuracy, allowing the characterization of complex centromere and telomere regions in nearly all chromosomes. Our high-quality genome assemblies also enable a comprehensive structural variation discovery among genomes of different pear species, leading to the identification of 6 341 high-confident SVs. Along with Brassicaceae (Dolatabadian et al., 2022) and soybean (Bayer et al., 2021), deletions in *Pco*, *Pdr* and *PsbF* genomes result in a complete loss of two copies of disease-resistance related genes, reflecting the differentiation of disease resistance in pear species. Hence, these two high-quality reference genomes provide valuable resources for detecting genomic variations in diverse pear species.

To capture the complete genetic diversity of pear, we constructed the pear pangenome using the *PsbM* reference genome as the backbone genome and identified 21 224 novel genes in the non-reference genome. By aligning the read sequencing data to the *PsbM* reference and the pear pangenome, we identified a

higher mapping rate, and a larger number of SNPs in the pan-genome, reflecting that the pan-genome can effectively reduce the single reference bias for mapping and variant calling. In contrast to SNP markers, our analysis based on PAVs provides additional insights into the population structure of pear, similar to *Arabidopsis* (Jiao and Schneeberger, 2020), *Amborella* (Hu et al., 2022) and rice (Wang et al., 2023a). Furthermore, we report distinct PAV occurrence frequencies and gene expression patterns in Asian and European populations for genes linked to pear fruit quality, including fruit shape, stone cells, aroma and fruit firmness. The *ADH* gene (*Pco_Presence_Seq_011843*) (Zhang et al., 2010; Shi et al., 2019) exhibits a higher frequent occurrence in European populations, alongside upregulated expression in European pear. These results are consistent with previous findings that SVs have significant impacts on fruit quality traits such as shape, flesh color, and sucrose content (Zhou et al., 2019; Song et al., 2020; Lyu et al., 2023). While the association between the PAV gene and gene expression levels remains unclear, additional research is required to elucidate the regulatory mechanisms of how PAV influences gene expression.

Using the constructed pear pan-genome, a 104 bp structural variation upstream of the *PsbMGH3.1* gene promoter was identified, contrasting with the corresponding position in *PsbFGH3.1*. This variation has the potential to augment gene expression responsible for sepal abscission, potentially leading to increased sepal shedding in 'Xuehuali'. Furthermore, we investigated the functional role of *PsbMGH3.1*, an auxin nicotinamide synthase gene, recognized as an early responder in the indole-3-acetic acid (IAA) signaling pathway, critical in plant organ shedding encompassing biosynthesis, transportation, and catabolism processes (Pencik et al., 2013; Qi et al., 2013; El Houari et al., 2023; Ding et al., 2024; Guo et al., 2024). IAA's binding to amino acids such as IAA-leucine, IAA-alanine, and IAA-phenylalanine through amide bonds, as well as its storage in a bound state with amino acids, indirectly reducing intracellular free IAA content, is highlighted. Our study predicted potential hydrogen bond formation between indole-3-acetic acid (IAA) and Tyr177 and Asp308 of *PsbMGH3.1*, indicating the likelihood of *PsbMGH3.1* interaction with IAA. Notably, *PsbMGH3.1* displayed upregulation in the detached zone of the paternal haplotype genomic cultivar 'Xuhuali', leading to a characteristic low-concentration auxin phenotype in heterologous overexpressing transgenic plants. *PsbMGH3.1* potentially influences pear sepal formation by modulating IAA content in plant cells, a critical aspect given the impact of persistent sepals on pear fruit quality and the significance of understanding sepal shedding at the molecular level for the cultivation of sepal-shedding fruits.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.hpj.2024.05.005>.

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